

EXP NO:2

Determination and plotting of Electric field and Electric flux density in sphere full of charges with varying radius.

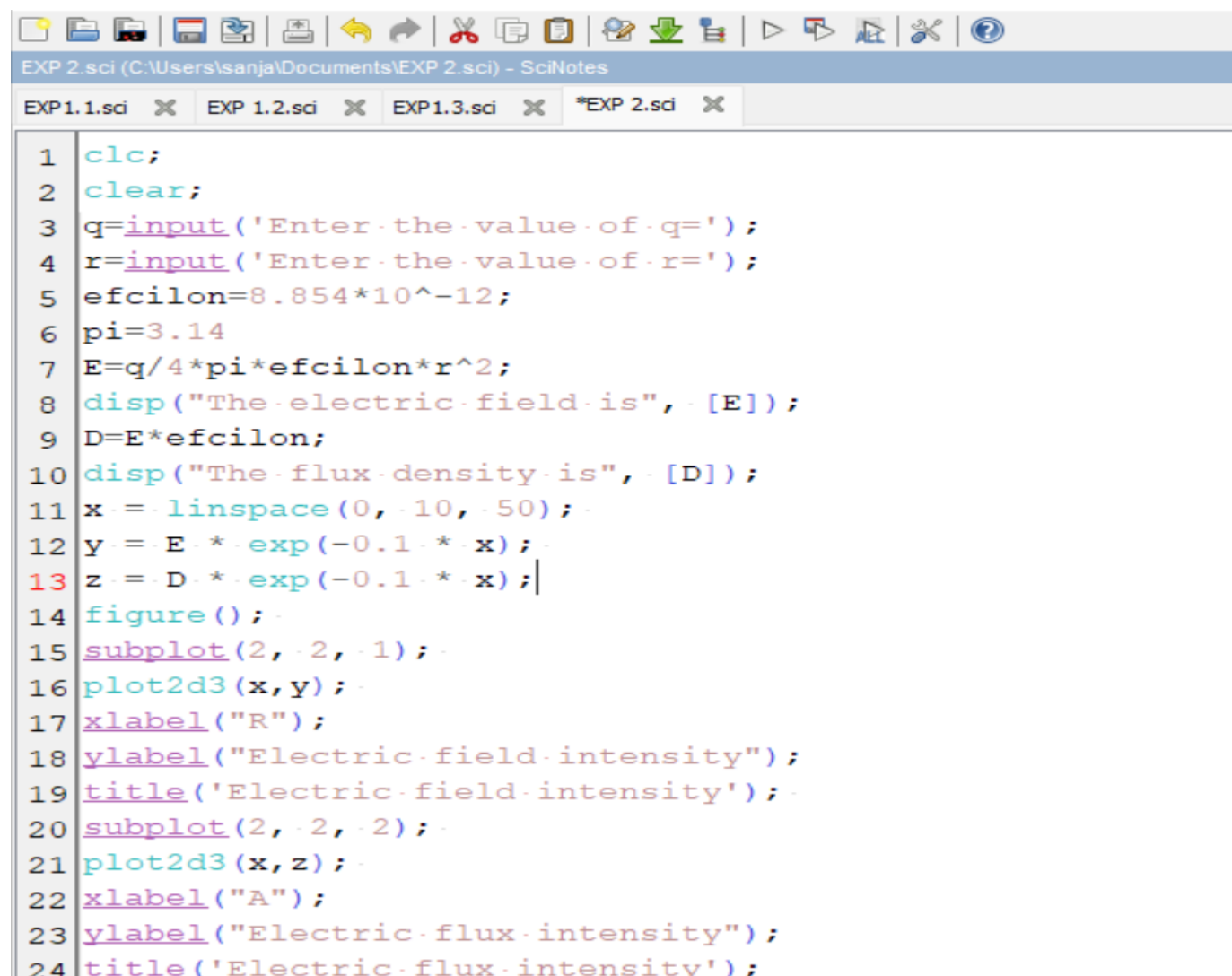
Aim:

To write program for electric field and electric flux density in sphere using SCILAB.

Software required:

- SCILAB version 6.0.1

PARTIAL CALUCATION:



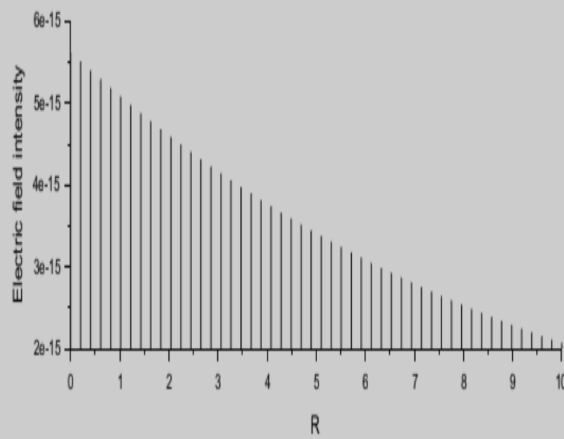
```
1 clc;
2 clear;
3 q=input('Enter the value of q=');
4 r=input('Enter the value of r=');
5 efcilon=8.854*10^-12;
6 pi=3.14
7 E=q/4*pi*efcilon*r^2;
8 disp("The electric field is", [E]);
9 D=E*efcilon;
10 disp("The flux density is", [D]);
11 x= linspace(0, 10, 50);
12 y= E * exp(-0.1 * x);
13 z= D * exp(-0.1 * x);
14 figure();
15 subplot(2, 2, 1);
16 plot2d3(x, y);
17 xlabel("R");
18 ylabel("Electric field intensity");
19 title('Electric field intensity');
20 subplot(2, 2, 2);
21 plot2d3(x, z);
22 xlabel("A");
23 ylabel("Electric flux intensity");
24 title('Electric flux intensity');
```

File Tools Edit ?

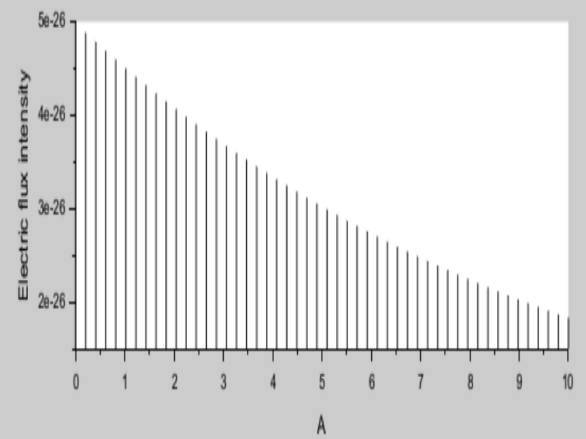


Graphic window number 0

Electric field intensity



Electric flux intensity



File Edit Control Applications ?



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- cntrlsys4.sci
- cntrlsys5.sci

Scilab 2025.1.0 Console

Enter the value of $q=10 \times 10^{-6}$

Enter the value of $r=9$

"The electric field is"

5.630D-15

"The flux density is"

4.985D-26

exec: Wrong number of output argument(s): 0 expected.

-->

THEROTICAL CALUCATION:

Q₂) i) Electric field intensity

$Q = 10 \times 10^{-6}$
 $R = 9 \text{ cm}$

$E = \frac{Q}{4\pi\epsilon_0 r^2}$

$D = \epsilon_0 E$

$F = \frac{Q_1 Q_2}{4\pi\epsilon_0 r^2}$

$E = \frac{Q}{4\pi\epsilon_0 r^2}$

$= \frac{10 \times 10^{-6}}{4(3.14) \times 8.854 \times 10^{-12} (9 \times 10^{-2})^2}$

$= 1110.1606 \text{ V/m}$

$\Rightarrow \text{Flux Density}$

$D = \epsilon_0 E$

$(8.854 \times 10^{-12})(1110.1606)$

$= 9.829 \text{ C/m}^2$

RESULT:

The electric field intensity (E) and electric flux density (D) of a uniformly charged sphere were successfully determined and plotted for different values of radius. The plots showed that the electric field and flux density vary with the distance from the centre of the sphere and change according to the radius of the charged sphere.